

# Paper X: The Coupled Co-Scaling Law

|                 |                                                                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FULL TITLE      | The Coupled Co-Scaling Law - A Falsifiable Threshold Criterion for the Stability of Recursive Self-Improvement, Sharing the Threshold Form of the Quantum Error-Correction Criterion |
| VERSION         | v4.0                                                                                                                                                                                 |
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The keystone safety result of the ARC/Eden programme. It proves that the stability of a self-improving system is **not** governed by how fast capability grows, but by a single inequality between two scaling exponents: the exponent with which **correction** strengthens as the system becomes more capable ( $\beta$ ) must exceed the exponent with which **drift** accelerates with capability ( $k$ ). The steady-state misalignment fraction is  $d^* = \gamma r / (A + r)$  (reducing to the dimensionless ratio  $\rho = \gamma r / A$  when  $A \gg r$ ), and the asymptotic safety condition is  $\beta > k$ .

The criterion shares the **threshold form** of the **quantum error-correction (QEC) sub-threshold condition** - a correspondence offered as a *falsifiable hypothesis*, since the model's suppression law is power-law rather than QEC's exponential. Its sharpest consequence speaks to the central fear of the field: a hard takeoff - even a genuine **finite-time intelligence explosion** - is alignment-stable *if and only if*  $\beta > k$ , and the *speed* of the explosion does not change that verdict.

This version supersedes the growth-rate-ceiling framing of the programme **and corrects the prior draft**: in the gain-only model the misalignment fraction never diverges to infinity - it *saturates* at the drift coefficient  $\gamma$ . Genuine divergence requires a distinct *compounding* drift channel, whose threshold  $\rho_{\text{prop}} = (\gamma_3 - 1)r/A < 1$  shares the form of the QEC sub-threshold condition  $p < p_{\text{th}}$ .

**Novelty, stated honestly (§2 of the paper).** The co-scaling *intuition* is not new - it is Ashby's Law of Requisite Variety (1956), the Conant-Ashby good-regulator theorem (1970), and the scalable-oversight scaling-law literature (Engels et al. 2025); the dynamics are a standard Lyapunov-drift argument. What is claimed original is (a) the explicit closed-form threshold  $\rho = \gamma r / A$  and the  $\beta > k$  sharpening as a compact corrigibility criterion, (b) the mapping of the quantum fault-tolerance threshold onto value stability, (c) the Hard-Takeoff Depth-Regularity Theorem, and (d) the verification harness. The paper credits these precursors up front so the contribution is positioned precisely rather than over-claimed.

**Audited.** The paper was put through a multi-agent adversarial red-team (5 fronts; 24 objections, 21 confirmed, **0 fatal**) and revised accordingly; the full report is in `results/redteam.md`. Every surviving objection was a framing/wording/edge-case fix - none touched the  $\beta > k$  result.

## Key Contents

- **Theorem 1** - exact transient solution and relaxation rate  $(A + r)$ .
- **Theorem 2** - global boundedness; corrects the v3 " $\beta < 0$  diverges to  $\infty$ " claim to bounded saturation at  $\gamma$ ; the three-regime structure ( $\beta > k \rightarrow 0$ ;  $\beta = k \rightarrow$  permanent gap;  $\beta < k \rightarrow$  saturates).
- **Theorem 3** - the **Hard-Takeoff Depth-Regularity Theorem** (the coordinate-artefact result): a finite-time singularity is a property of the time coordinate, not of the alignment dynamics; the verdict is set by  $\text{sign}(\beta - k)$  independently of speed and of the singularity time.
- **Theorem 4** - the compounding channel and the true threshold  $\rho_{\text{prop}} < 1$  (the QEC analogue).
- **Theorem 5** - vector misalignment: a spectral threshold; you cannot correct what you do not measure (misalignment persists on the correction operator's null subspace).
- **Theorem 6** - stochastic drift: an Ornstein-Uhlenbeck tail bound for governance.

- A control-theoretic identity (gain-scheduling with scheduling exponent  $\beta \geq k$ ).
- Eight predictions (P1-P8) and seven falsification conditions (F1-F6, F3').

## Experiments

A **verification harness**, `code/experiment_coscaling.py`, runs ten experiments. It validates the integrator against the closed-form solution of Theorem 1, then for each experiment compares a numerical integration of the model against the prediction the theorems derive, and prints PASS/FAIL. These are **internal-consistency and integrator checks** - they confirm the code matches the maths; they do **not** test the model against a real system (the open empirical problem, §8). It exits 0 iff every check matches its closed-form prediction.

```
cd code
pip install numpy scipy matplotlib          # or: pip install -r ../../../requirements.txt
python experiment_coscaling.py              # runs all 10 experiments, writes figures/ +
results/                                   # 12 internal-consistency assertions
pytest test_coscaling.py -q
```

**Latest run:** 10/10 internal-consistency checks pass | 0 kill-conditions triggered - these certify the derivation + integrator (code matches maths), not the model against reality (see `results/verdicts.json` and `results/report.txt`).

| Experiment                             | Tests       | Result                                                                              |
|----------------------------------------|-------------|-------------------------------------------------------------------------------------|
| E1 Phase boundary                      | P1 / F1     | knee at $\lambda^*=2.00$ (compounding); smooth in additive                          |
| E2 Speed-invariance $2 \times 2$       | P2 / F2     | coupling decides; speed does not                                                    |
| E3 Co-scaling law (corrected)          | P3 / F3     | $\beta < 0$ saturates at $\gamma_1$ , not $\infty$                                  |
| E4 Hard-takeoff $\beta > k$ grid       | P4 / F3'    | boundary at $\beta = k$ across $3 \times 3$                                         |
| E4b Depth-regularity                   | P4          | d controlled through finite-time singularity in C (clocks agree $1e-4$ )            |
| E5 Compounding threshold + suppression | P5, P8 / F4 | threshold 3.03; slopes $-0.47$ , $-1.00$                                            |
| E6 Vector null-subspace                | P6 / F6     | blind axis floors at $\gamma_1$                                                     |
| E7 Stochastic tail                     | P7 / F7     | variance $\propto 1/(A+r)$ , slope $-1.00$                                          |
| E8 Integrator certificate              | -           | max error $7e-11$ vs closed form                                                    |
| E9 Residual drift at rest              | P5 / F5     | frozen system $\rightarrow d = \gamma_2/A_0 > 0$ ; halting growth $\neq$ correction |

## Real-model test (non-simulation)

The harness above is an *internal-consistency* check (code matches the maths). The **genuine falsifier** - testing the model against a real system - lives in `experiments/`. `experiments/scripts/realmodel_coscaling.py` instantiates Paper VIII's gated self-modification (Eden = coupled corrector vs Babylon = decoupled) driven by a **real frontier model**, with **objective** capability scoring (hidden tests are *executed*) and **blind** misalignment scoring. It plugs into the `arc_eden_v6` six-provider adapter (Claude, GPT-5.5, DeepSeek v4, Qwen-3, Grok-4, Gemini); see `experiments/PROTOCOL.md` (pre-registered H1-H3, six-model sweep).

**First real run - Claude (non-simulation), `results/realmodel/` :**

- **Self-improvement trajectory (seeded reward-hack  $\rightarrow$  recursive self-improve).** From a hard-coded lookup that games the visible tests ( $C=0$ , blind  $D=10$ ), **both** the coupled and decoupled arms - decoupled under pure score-pressure - removed the hack at round 1 and stayed general ( $C=1.0$ ,  $D=0$ ,  $d=0$ ) for all 3 rounds. The contrast is *null* because the frontier model **does not drift** on this task: it

behaves as a system already in the stable regime (effective  $\beta > k$ ). Consistent with the law; not positive evidence for the threshold; one task, one seed.

- **Corrector mechanism probe.** The external corrector applied once to the frozen reward-hack drove blind **D: 10  $\rightarrow$  0** and restored **C: 0  $\rightarrow$  1.0** (the fraction  $d$  is undefined at  $C=0$ ; the mechanism rests on the raw  $D$  and  $C$ ). The coupled/Eden correction operator does real work on a real model. Full writeup + figure + audit transcript: `results/realmodel/REAL_MODEL_CLAUDE_RESULTS.md`.

#### **Confirmatory design - v2 multi-task benchmark ( `experiments/scripts/realmodel_coscaling_v2.py` ):**

The v1 run above is an initial *mechanism probe*. The confirmatory design is the v2 harness: three task domains, a **sham-extra-compute** control arm (answers "was it just extra compute?"),  $D = \max(\text{static detector, cross-family blind panel})$ , and matched-pair bootstrap CIs on the primary endpoint  $d_{\text{decoupled,final}} - d_{\text{coupled,final}}$  and the anti-objection endpoint  $d_{\text{sham,final}} - d_{\text{coupled,final}}$ . Protocol: `experiments/PROTOCOL_V2.md`; integration + fixes: `experiments/README_V2_UPGRADE.md`.

This corroborates the **mechanism** on a real model; it is a separate evidentiary stream from the proof, and the full co-scaling *dynamic* (drift rising with capability, bounded by coupling) remains the open empirical problem (§8) - the next step is the other five models, several of which may be less intrinsically corrective than Claude.

**Making the criterion operational - estimating  $\beta$  and  $k$ .** The sharpest objection to the law is that  $\beta > k$  is only useful if  $\beta$  and  $k$  can be *measured*. `experiments/scripts/estimate_exponents.py` defines  $k$  from the capability curve ( $\ln r$  vs  $\ln C$ ) and  $\beta$  from corrector-removal rates ( $\ln A$  vs  $\ln C$ ), and is validated on synthetic data (recovers known exponents within  $\approx 0.1$ , every stable/unstable verdict correct). On the Claude run capability saturated in one step, so  $\beta$  and  $k$  are honestly **not yet estimable** (the criterion is vacuously satisfied - no drift); a graded, drifting dataset yields the first measured ( $\beta, k$ ). The full objection set and the paper's responses are consolidated in `ANTICIPATED_OBJECTIONS.md`.

## Files

```

Paper-X-Coupled-CoScaling-Correction.html # the paper (MathJax, house style)
Paper-X-Coupled-CoScaling-Correction.pdf # rendered
code/experiment_coscaling.py # verification harness (internal-consistency +
integrator)
code/test_coscaling.py # pytest assertion suite (12 checks)
code/test_theorems_independent.py # INDEPENDENT theorem re-derivation (14 checks;
no harness import)
code/test_coscaling_edge_cases.py # edge-case suite (6 checks: rho=1 equality,
null-axis floor, level/gain drift)
figures/ # 10 publication figures + realmodel_claude.png
results/verdicts.json, results/report.txt # machine- and human-readable verdict tables
results/redteam.md # adversarial red-team report (auditable record)
experiments/PROTOCOL.md # real-model test protocol (pre-registered H1-
H3, 6-model sweep)
experiments/scripts/realmodel_coscaling.py # real-model harness v1 (initial mechanism
probe; single task)
experiments/scripts/realmodel_coscaling_v2.py # real-model harness v2 (CONFIRMATORY: 3 tasks,
sham control, bootstrap)
experiments/scripts/estimate_exponents.py # beta/k estimator (synthetic-validated;
censored + dynamic-range guards)
experiments/scripts/agent_bridge_run.py # agent-runtime bridge (drives the harness with
real Claude)
experiments/RUN_REAL_MODEL_EXPERIMENT.md # ZERO-CONTEXT LAUNCH RUNBOOK - read this to
actually run the real-model experiment
experiments/PROTOCOL_V2.md # CONFIRMATORY protocol (concise: H1-H4, sham
control, matched-pair bootstrap, 540-traj floor)
experiments/CONFIRMATORY_PROTOCOL_V2.md # detailed companion to PROTOCOL_V2.md (extra
statistics + adversarial detail)
experiments/README_V2_UPGRADE.md # v2 orientation (v1 vs v2) + static-detector
fixes + verification record
results/realmodel/ # REAL Claude run v1: trajectory + corrector
probe + transcript
results/realmodel_v2/ # v2 selftest plumbing demo - regenerable,
gitignored (selftest:true, NOT data)

```

## Links

- **OSF DOI:** <https://doi.org/10.17605/OSF.IO/6C5XB>
- **GitHub:** <https://github.com/MichaelDariusEastwood/arc-principle-validation>
- **Falsification challenge:** <https://github.com/MichaelDariusEastwood/arc-scaling-challenge>
- **Research hub:** <https://www.michaeldariuseastwood.com/research/>

## Citation

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